

Nonlinear Dynamics of Electron-Beam-Driven CO Molecules in an Active Optical Cavity

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Abstract

In this literature we derive the first draft of a seven-dimensional non-autonomous dynamical system for the coupled vibro-plasma-cavity-inversion dynamics of CO molecules under active discharge conditions. The model follows the coupled evolution of molecular vibration, plasma density, electron energy, vibrational inversion, and the cavity field under pulsed electron-beam excitation.

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1 Notation and Variable Definitions

1.1 Physical Variables

Symbol	Definition	Units
r	Internuclear separation	m
r_e	Equilibrium bond length	m
E_{phys}	Cavity electric field amplitude	V m ⁻¹
n_e	Free electron number density	m ⁻³
ε_{phys}	Mean electron kinetic energy	J
n_{inv}	Vibrationally excited CO density	m ⁻³
t	Physical time	s

1.2 Molecular and Discharge Parameters

1.3 Characteristic Scales

$$\omega_M = \sqrt{\frac{2a^2 D_e}{m}}, \quad \text{molecular vibrational frequency (rad s}^{-1}\text{)}, \quad (1)$$

$$E_* = \frac{2aD_e}{\mu_1}, \quad \text{characteristic field scale (V m}^{-1}\text{)}, \quad (2)$$

$$\varepsilon_{ref} = \varepsilon_I, \quad \text{reference electron energy (J)}, \quad (3)$$

$$n_{ref} = \frac{\delta}{\beta_r}, \quad \text{reference electron density (m}^{-3}\text{)}. \quad (4)$$

1.4 Dimensionless Parameters

$$b' = \frac{b}{\omega_M}, \quad \gamma = \frac{\kappa}{\omega_M}, \quad \Omega_c = \frac{\omega_c}{\omega_M}, \quad g' = \frac{g}{\omega_M}, \quad (5)$$

$$\chi = \frac{n_{ref} e^2}{\epsilon_0 m_e \omega_M^2}, \quad \delta' = \frac{\delta}{\omega_M}, \quad \beta'_r = \frac{\beta_r n_{ref}}{\omega_M}, \quad \mu_c = b' g'^2, \quad (6)$$

$$H_{cav} = \frac{\alpha_{cav} E_*^2}{\omega_M \varepsilon_I}, \quad H_{DC} = \frac{\alpha_{DC} E_{DC}^2}{\omega_M \varepsilon_I}, \quad H_{beam} = \frac{P_{beam}}{\varepsilon_I n_{ref} \omega_M}, \quad \Gamma_\varepsilon = \frac{1}{\omega_M \tau_\varepsilon}, \quad (7)$$

$$\varepsilon'_0 = \frac{\varepsilon_0}{\varepsilon_I}, \quad K'_0 = \frac{K_0 n_g}{\omega_M}, \quad K'_{EV} = \frac{K_{EV} n_g}{\omega_M}, \quad \varepsilon'_{EV} = \frac{\varepsilon_{EV}}{\varepsilon_I}, \quad (8)$$

$$\Gamma'_N = \frac{\Gamma_N}{\omega_M}, \quad \sigma' = \frac{\sigma E_*^2}{\omega_M}, \quad \kappa'_g = \frac{G_\kappa}{\omega_M^2}, \quad G_0 = \frac{\kappa'_g}{\Gamma'_N}, \quad \alpha_s = \frac{\sigma'}{\Gamma'_N}, \quad (9)$$

where $g^2 = N_{mol} \mu_1^2 / (\epsilon_0 V m)$ is the collective coupling frequency squared, $\alpha_{cav} = e^2 \nu_c / (2m_e \omega_M^2)$ and $\alpha_{DC} = e^2 / (2m_e \nu_c)$ are the Drude heating coefficients derived in Section 4, and κ'_g is a phenomenological coupling coefficient discussed in Section 11.

2 Morse Oscillator and Molecular Equation of Motion

2.1 Morse Potential and Force Function

The Morse potential for a diatomic molecule with reduced mass m is:

$$V_M(r) = D_e \left(1 - e^{-a(r-r_e)} \right)^2. \quad (10)$$

Symbol	Definition	Units
D_e	Morse dissociation energy	J
a	Morse range parameter	m^{-1}
m	Reduced mass of CO	kg
μ_1	Dipole moment derivative	C
b	Molecular damping coefficient	s^{-1}
e	Elementary charge	C
m_e	Electron mass	kg
ν_c	Electron–neutral collision frequency	s^{-1}
ε_I	CO ionisation energy	J
ε_{EV}	CO vibrational excitation threshold energy	J
δ	Electron loss rate	s^{-1}
β_r	Recombination coefficient	$\text{m}^3 \text{s}^{-1}$
K_0	Ionisation rate coefficient (peak)	$\text{m}^3 \text{s}^{-1}$
K_{EV}	Vibrational excitation rate coefficient (peak)	$\text{m}^3 \text{s}^{-1}$
n_g	Neutral CO number density	m^{-3}
N_{mol}	Number of CO molecules in cavity	—
V	Cavity mode volume	m^3
Γ_N	Vibrational relaxation rate	s^{-1}
σ	Stimulated emission cross-section	$\text{m}^2 \text{V}^{-2} \text{s}^{-1}$
G_κ	Inversion-mediated cavity coupling coefficient	s^{-2}
E_{DC}	DC discharge electric field	V m^{-1}
τ_ε	Electron energy relaxation time	s
S_{beam}^{phys}	Volumetric beam electron source rate	$\text{m}^{-3} \text{s}^{-1}$
P_{beam}	Beam power density deposited in gas	W m^{-3}
ω_c	Cavity resonance frequency	rad s^{-1}
κ	Cavity field decay rate	s^{-1}
ε_0	Thermal equilibrium electron energy	J
ϵ_0	Permittivity of free space	F m^{-1}
η_{bg}	Background electron density (dimensionless)	—
S_{DC}	DC electron source term (dimensionless)	—
T	Pulse repetition period (dimensionless)	—
N_p	Number of pulses in train	—

The restoring force is:

$$F_{Morse} = -\frac{dV_M}{dr} = -\frac{d}{dr} \left[D_e \left(1 - e^{-a(r-r_e)} \right)^2 \right]. \quad (11)$$

Using the chain rule:

$$\begin{aligned} \frac{dV_M}{dr} &= D_e \cdot 2 \left(1 - e^{-a(r-r_e)} \right) \cdot \frac{d}{dr} \left(1 - e^{-a(r-r_e)} \right) \\ &= 2D_e \left(1 - e^{-a(r-r_e)} \right) \cdot \left(a e^{-a(r-r_e)} \right) \\ &= 2aD_e \left(1 - e^{-a(r-r_e)} \right) e^{-a(r-r_e)}. \end{aligned} \quad (12)$$

Therefore:

$$F_{Morse} = -2aD_e \left(1 - e^{-a(r-r_e)} \right) e^{-a(r-r_e)}. \quad (13)$$

Introducing $x = a(r - r_e)$ (the dimensionless bond stretch) and the Morse force function:

$$S(x) = e^{-x} - e^{-2x}, \quad (14)$$

we obtain $F_{Morse} = -2aD_e S(x)$.

Checking this against the previous line:

$$\left(1 - e^{-a(r-r_e)}\right) e^{-a(r-r_e)} = (1 - e^{-x}) e^{-x} = e^{-x} - e^{-2x} = S(x). \quad \checkmark \quad (15)$$

Taylor-expanding near equilibrium, with $e^{-x} = 1 - x + x^2/2 - x^3/6 + \dots$ and $e^{-2x} = 1 - 2x + 2x^2 - 4x^3/3 + \dots$:

$$\begin{aligned} S(x) &= \left(1 - x + \frac{x^2}{2} - \frac{x^3}{6}\right) - \left(1 - 2x + 2x^2 - \frac{4x^3}{3}\right) + O(x^4) \\ &= x - \frac{3x^2}{2} + \frac{7x^3}{6} + O(x^4). \end{aligned} \quad (16)$$

The linear term x gives harmonic behaviour; the x^2 term is the leading anharmonic correction.

2.2 Dimensional Equation of Motion

For one CO molecule coupled to the cavity field E_{phys} with molecular damping b :

$$m\ddot{r} = -2aD_e S(x) - mbr + \mu_1 E_{phys}. \quad (17)$$

Dividing by m :

$$\ddot{r} = -\frac{2aD_e}{m} S(x) - br + \frac{\mu_1}{m} E_{phys}. \quad (18)$$

2.3 Non-dimensionalisation

With $\tau = \omega_M t$ (dimensionless time), $x = a(r - r_e)$ (bond stretch), $y = (a/\omega_M)\dot{r}$ (dimensionless bond velocity), $\mathcal{E} = E_{phys}/E_*$ (dimensionless cavity field), we have:

$$\dot{r} = \frac{y\omega_M}{a}, \quad (19)$$

$$\ddot{r} = \frac{d}{dt} \left(\frac{y\omega_M}{a} \right) = \frac{\omega_M}{a} \frac{dy}{dt} = \frac{\omega_M}{a} \cdot \omega_M \frac{dy}{d\tau} = \frac{\omega_M^2}{a} \frac{dy}{d\tau}. \quad (20)$$

Substituting into (18) and multiplying through by a/ω_M^2 :

$$\frac{a}{\omega_M^2} \cdot \frac{\omega_M^2}{a} \frac{dy}{d\tau} = -\frac{a}{\omega_M^2} \cdot \frac{2aD_e}{m} S(x) - \frac{a}{\omega_M^2} \cdot b \cdot \frac{y\omega_M}{a} + \frac{a}{\omega_M^2} \cdot \frac{\mu_1}{m} E_* \mathcal{E}. \quad (21)$$

Simplifying:

$$\frac{dy}{d\tau} = -\frac{2a^2D_e}{m\omega_M^2} S(x) - \frac{b}{\omega_M} y + \frac{a\mu_1 E_*}{m\omega_M^2} \mathcal{E}. \quad (22)$$

The three coefficients work out as follows.

First coefficient:

$$\frac{2a^2D_e}{m\omega_M^2} = \frac{2a^2D_e}{m \cdot 2a^2D_e/m} = \frac{2a^2D_e}{2a^2D_e} = 1. \quad (23)$$

Second coefficient, where $b' = b/\omega_M$ is the dimensionless molecular damping rate:

$$\frac{b}{\omega_M} = b'. \quad (24)$$

Third coefficient:

$$\frac{a\mu_1 E_*}{m\omega_M^2} = \frac{a\mu_1 \cdot 2aD_e/\mu_1}{m \cdot 2a^2D_e/m} = \frac{2a^2D_e}{2a^2D_e} = 1. \quad (25)$$

The dimensionless molecular subsystem is therefore:

$$\boxed{x' = y, \quad y' = -S(x) - b'y + \mathcal{E}.} \quad (26)$$

3 Cavity Field Equation with Saturating Plasma Response

3.1 Projected Maxwell–Ampere Equation

For a single resonant cavity mode, the projected field equation is:

$$\ddot{E}_{phys} + 2\kappa\dot{E}_{phys} + \omega_c^2 E_{phys} = -\frac{1}{\epsilon_0}\ddot{P} - \frac{1}{\epsilon_0}\dot{J}_{plasma} + G_\kappa N E_{phys}, \quad (27)$$

where κ is the cavity field decay rate, ω_c is the bare cavity resonance frequency, P is the molecular polarisation, J_{plasma} is the free-electron current density, and G_κ is the (dimensional) inversion-mediated cavity coupling coefficient introduced in Section 11.

3.2 Molecular Polarisation Source

The total polarisation from N_{mol} molecules (with cavity mode volume V) is:

$$P = \frac{N_{mol}\mu_1}{V}(r - r_e) = \frac{N_{mol}\mu_1}{aV}x, \quad (28)$$

so $\ddot{P} = (N_{mol}\mu_1/V)\ddot{r}$.

Substituting the equation of motion (17):

$$\begin{aligned} -\frac{1}{\epsilon_0}\ddot{P} &= -\frac{N_{mol}\mu_1}{V\epsilon_0}\ddot{r} \\ &= -\frac{N_{mol}\mu_1}{V\epsilon_0 m}(-2aD_e S(x) - mbr\dot{r} + \mu_1 E_{phys}) \\ &= \frac{2aD_e N_{mol}\mu_1}{\epsilon_0 V m}S(x) + \frac{bN_{mol}\mu_1}{\epsilon_0 V}\dot{r} - \frac{N_{mol}\mu_1^2}{\epsilon_0 V m}E_{phys}. \end{aligned} \quad (29)$$

The collective coupling frequency is defined as:

$$g^2 = \frac{N_{mol}\mu_1^2}{\epsilon_0 V m}, \quad [g^2] = \text{rad}^2 \text{s}^{-2}. \quad (30)$$

Using g^2 and $E_* = 2aD_e/\mu_1$:

$$\frac{2aD_e N_{mol}\mu_1}{\epsilon_0 V m} = \frac{2aD_e}{\mu_1} \cdot \frac{N_{mol}\mu_1^2}{\epsilon_0 V m} = E_* g^2. \quad (31)$$

Therefore:

$$-\frac{1}{\epsilon_0}\ddot{P} = E_* g^2 S(x) + \frac{bN_{mol}\mu_1}{\epsilon_0 V}\dot{r} - g^2 E_{phys}. \quad (32)$$

3.3 Plasma Current: Linear Drude Model

For free electrons with density n_e , charge e , and mass m_e , in the limit $\omega_M \gg \nu_c$ (where ν_c is the electron–neutral collision frequency), Newton’s law gives $m_e \dot{v}_e = eE_{phys}$, so:

$$\dot{J}_{plasma} = n_e e \dot{v}_e = n_e e \cdot \frac{eE_{phys}}{m_e} = \frac{n_e e^2}{m_e} E_{phys} = \epsilon_0 \varpi^2 E_{phys}, \quad (33)$$

where

$$\varpi^2(n_e) = \frac{n_e e^2}{\epsilon_0 m_e} \quad (34)$$

is the (linear) plasma frequency associated with the free-electron density n_e .

Therefore:

$$-\frac{1}{\epsilon_0}\dot{J}_{plasma} = -\varpi^2(n_e)E_{phys}. \quad (35)$$

3.4 Physical Limitation and Saturating Response

The linear Drude response $\varpi^2 \propto n_e$ only holds at low plasma densities. As n_e increases, several things break this assumption at once: the plasma frequency starts to saturate, ionisation depletes the neutral gas, the dielectric response becomes nonlinear, and the mode structure itself changes.

The effective cavity frequency from the linear model is:

$$\omega_{eff,linear}^2 = \omega_c^2 + g^2 - \varpi^2(n_e). \quad (36)$$

This becomes unphysical when:

$$\varpi^2(n_e) > \omega_c^2 + g^2 \quad \Rightarrow \quad \omega_{eff,linear}^2 < 0, \quad (37)$$

turning the cavity into an inverted oscillator with exponential growth.

To model saturation physically, we introduce a saturation density scale n_{sat} and its dimensionless counterpart $\eta_{sat} = n_{sat}/n_{ref}$:

$$\boxed{\varpi_{eff}^2(n_e) = \varpi_{sat}^2 \frac{n_e/n_{sat}}{1 + n_e/n_{sat}} = \frac{\varpi_{sat}^2 \eta/\eta_{sat}}{1 + \eta/\eta_{sat}} = \frac{\chi\eta}{1 + \eta/\eta_{sat}}}, \quad (38)$$

where $\chi = n_{ref}e^2/(\epsilon_0 m_e \omega_M^2)$ as before and $\varpi_{sat}^2 = \chi \omega_M^2$. The parameter η_{sat} is a free, physically motivated saturation parameter: it sets the dimensionless electron density at which the plasma response begins to depart from the linear Drude regime and saturate. It is specified as an input parameter of the model, subject only to the stability requirement derived below (Eq. 44).

In the two limits:

$$\text{For } \eta \ll \eta_{sat} : \quad \varpi_{eff}^2 \approx \chi\eta \quad (\text{linear regime}), \quad (39)$$

$$\text{For } \eta \gg \eta_{sat} : \quad \varpi_{eff}^2 \rightarrow \chi\eta_{sat} \quad (\text{saturated}). \quad (40)$$

3.5 Effective Cavity Frequency with Saturation

Substituting (32) and the saturating plasma term into (27), and moving $-g^2 E_{phys}$ to the left:

$$\ddot{E}_{phys} + 2\kappa\dot{E}_{phys} + (\omega_c^2 + g^2 - \varpi_{eff}^2(n_e)) E_{phys} = E_* g^2 S(x) + \frac{bN_{mol}\mu_1}{\epsilon_0 V} \dot{r} + G_\kappa N E_{phys}. \quad (41)$$

The effective resonance frequency is:

$$\boxed{\omega_{eff}^2(n_e) = \omega_c^2 + g^2 - \varpi_{eff}^2(n_e) = \omega_c^2 + g^2 - \frac{\chi\eta}{1 + \eta/\eta_{sat}}}. \quad (42)$$

The plasma shift is bounded:

$$\frac{\chi\eta}{1 + \eta/\eta_{sat}} \leq \chi\eta_{sat} \quad \text{for all } \eta \geq 0. \quad (43)$$

To keep the system stable for all physically accessible densities, η_{sat} must satisfy:

$$\chi\eta_{sat} < \omega_c^2 + g^2, \quad \text{equivalently} \quad \eta_{sat} < \frac{\Omega_c^2 + g'^2}{\chi}. \quad (44)$$

Any η_{sat} satisfying (44) guarantees $\omega_{eff}^2(\eta) > 0$ for all $\eta \geq 0$, so the cavity can never be driven into the unphysical inverted-oscillator regime of Eq. (37). Throughout the rest of this paper η_{sat} is treated as a specified model input satisfying this bound, rather than as a quantity derived from a particular numerical instability threshold.

3.6 Derivation of μ_c

Convert $\dot{r} = y\omega_M/a$. Dividing (41) through by $\omega_M^2 E_*$ to obtain the dimensionless $\mathcal{Y}'$ equation, the damping source term contributes:

$$\frac{1}{\omega_M^2 E_*} \cdot \frac{bN_{mol}\mu_1}{\epsilon_0 V} \cdot \frac{y\omega_M}{a} = \underbrace{\frac{bN_{mol}\mu_1}{\epsilon_0 V a \omega_M E_*}}_{\mu_c} \cdot y. \quad (45)$$

Evaluating μ_c step by step:

$$\begin{aligned} \mu_c &= \frac{bN_{mol}\mu_1}{\epsilon_0 V a \omega_M E_*} \\ &= \frac{b}{\omega_M} \cdot \frac{N_{mol}\mu_1}{\epsilon_0 V a E_*} \\ &= b' \cdot \frac{N_{mol}\mu_1}{\epsilon_0 V} \cdot \frac{1}{a E_*}. \end{aligned} \quad (46)$$

Using $N_{mol}\mu_1/(\epsilon_0 V) = g^2 m/\mu_1$ (from rearranging (30)):

$$\mu_c = b' \cdot \frac{g^2 m}{\mu_1} \cdot \frac{1}{a E_*}. \quad (47)$$

Using $E_* = 2aD_e/\mu_1$, so $\mu_1 a E_* = 2a^2 D_e = m\omega_M^2$ (from (1)):

$$\begin{aligned} \mu_c &= b' \cdot \frac{g^2 m}{\mu_1 a E_*} = b' \cdot \frac{g^2 m}{m\omega_M^2} \\ &= b' \cdot \frac{g^2}{\omega_M^2} = b' \cdot g'^2. \end{aligned} \quad (48)$$

$$\boxed{\mu_c = b' g'^2 = \frac{b}{\omega_M} \cdot \frac{g^2}{\omega_M^2}}. \quad (49)$$

Dimensional check: $[b'] = [b/\omega_M] = \text{s}^{-1}/\text{s}^{-1} = 1$, $[g'^2] = [g^2/\omega_M^2] = 1$, so $[\mu_c] = 1$. ✓

3.7 Dimensionless Cavity Equation

Dividing (41) by $\omega_M^2 E_*$ and using $\mathcal{Y}' = \ddot{E}_{phys}/(\omega_M^2 E_*)$, $\mathcal{Y} = \dot{E}_{phys}/(\omega_M E_*)$, $\mathcal{E} = E_{phys}/E_*$, and writing $\gamma = \kappa/\omega_M$, $\Omega_c = \omega_c/\omega_M$, $g' = g/\omega_M$, $\kappa'_g = G_\kappa/\omega_M^2$:

$$\mathcal{Y}' + 2\frac{\kappa}{\omega_M}\mathcal{Y} + \frac{\omega_c^2 + g^2 - \chi\eta/(1 + \eta/\eta_{sat})}{\omega_M^2}\mathcal{E} = \frac{g^2}{\omega_M^2}S(x) + \mu_c y + \frac{G_\kappa}{\omega_M^2}N\mathcal{E}. \quad (50)$$

Rearranging:

$$\mathcal{Y}' = -2\gamma\mathcal{Y} - \Omega_{eff}^2(\eta)\mathcal{E} + g'^2 S(x) + \mu_c y + \kappa'_g N\mathcal{E}, \quad (51)$$

where

$$\boxed{\Omega_{eff}^2(\eta) = \Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{sat}}}. \quad (52)$$

The term $\kappa'_g N\mathcal{E}$ in equation (51) is a gain-like term that acts multiplicatively on the field. Rearranging the equation makes its role clearer:

$$\mathcal{Y}' = -2\gamma\mathcal{Y} - (\Omega_{eff}^2(\eta) - \kappa'_g N)\mathcal{E} + g'^2 S(x) + \mu_c y. \quad (53)$$

So the vibrational inversion N acts as an inversion-mediated cavity coupling: it corrects the effective resonance frequency rather than amplifying the field's amplitude on its own. We define the inversion-corrected effective cavity resonance

$$\boxed{\Omega_{\text{cav}}^2(\tau) = \Omega_{\text{eff}}^2(\eta(\tau)) - \kappa'_g N(\tau)}, \quad (54)$$

which is the quantity that governs the coupled dynamics and that we track throughout the rest of the paper.

3.8 Seed Term and the Trivial Null Solution

Equation (51) has an important structural property: every term on the right-hand side is proportional to $\mathcal{E}$, $\mathcal{Y}$, y , or $S(x)$. If the cavity starts empty, $\mathcal{E}(0) = \mathcal{Y}(0) = 0$, and the molecular subsystem starts at rest at equilibrium, $x(0) = y(0) = 0$ so that $S(0) = 0$, then every source term coupled to the field vanishes identically at $\tau = 0$ and stays zero for all τ :

$$\mathcal{E}(\tau) \equiv 0, \quad \mathcal{Y}(\tau) \equiv 0 \quad \text{is an exact solution.} \quad (55)$$

Nothing in the deterministic system (51) can start the cavity oscillating from this trivial fixed point, no matter how large the gain coefficients g'^2 or κ'_g become: a homogeneous linear (in $\mathcal{E}$) system cannot amplify a signal that is not there.

Of course, no real cavity is ever perfectly empty – vacuum field fluctuations, thermal photons, spontaneous emission from the inversion, or electronic noise in the readout chain all provide some minute initial disturbance. These stochastic effects sit outside a deterministic ODE model, but their role – giving the gain mechanisms something to act on – still needs to be represented somehow if the model is to show any field buildup at all. We therefore add a small constant seed term E_{seed} to the field equation:

$$\mathcal{Y}' = -2\gamma\mathcal{Y} - \Omega_{\text{eff}}^2(\eta)\mathcal{E} + g'^2 S(x) + \mu_c y + \kappa'_g N\mathcal{E} + E_{\text{seed}}. \quad (56)$$

E_{seed} is a small, fixed, dimensionless constant (in the same units as $\mathcal{Y}'$) standing in for the vacuum/thermal fluctuation floor: it breaks the trivial fixed point $(\mathcal{E}, \mathcal{Y}) = (0, 0)$ so that the gain terms $g'^2 S(x)$, $\mu_c y$, and $\kappa'_g N\mathcal{E}$ have something non-zero to amplify once the electron energy and inversion have built up. Because E_{seed} is chosen to be numerically negligible next to the dynamically generated field once the system is above threshold, it does not change the physical content of the model – it only regularises the initial-value problem by replacing an unmodelled stochastic seed with an equivalent small deterministic one.

4 Drude Heating and Energy Balance

4.1 Cavity Field Heating

An electron in the oscillating cavity field $E_{\text{phys}}(t) = E_0 \cos(\omega_M t)$ satisfies:

$$m_e \dot{v} = eE_0 \cos(\omega_M t) - m_e \nu_c v. \quad (57)$$

Seeking steady-state solution $v(t) = \text{Re}[v_0 e^{-i\omega_M t}]$ and substituting:

$$\begin{aligned} m_e \frac{d}{dt} [\text{Re}(v_0 e^{-i\omega_M t})] &= eE_0 \cos(\omega_M t) - m_e \nu_c \text{Re}(v_0 e^{-i\omega_M t}), \\ m_e \text{Re}(-i\omega_M v_0 e^{-i\omega_M t}) &= e \text{Re}(E_0 e^{-i\omega_M t}) - m_e \nu_c \text{Re}(v_0 e^{-i\omega_M t}). \end{aligned} \quad (58)$$

Equating complex amplitudes:

$$\begin{aligned}
-im_e\omega_M v_0 &= eE_0 - m_e\nu_c v_0, \\
v_0(m_e\nu_c - im_e\omega_M) &= eE_0, \\
v_0 &= \frac{eE_0}{m_e(\nu_c - i\omega_M)}.
\end{aligned} \tag{59}$$

The time-averaged power absorbed per electron is:

$$\begin{aligned}
P_{abs}^{(cav)} &= \frac{1}{2} \text{Re}[eE_0 \cdot v_0^*] \\
&= \frac{1}{2} \text{Re} \left[eE_0 \cdot \frac{eE_0}{m_e(\nu_c + i\omega_M)} \right] \\
&= \frac{e^2 E_0^2}{2m_e} \text{Re} \left[\frac{1}{\nu_c + i\omega_M} \right] \\
&= \frac{e^2 E_0^2}{2m_e} \cdot \frac{\nu_c}{\nu_c^2 + \omega_M^2}.
\end{aligned} \tag{60}$$

For $\omega_M \gg \nu_c$:

$$P_{abs}^{(cav)} \approx \frac{e^2 \nu_c}{2m_e \omega_M^2} E_0^2 \equiv \alpha_{cav} E_{phys}^2, \quad \alpha_{cav} = \frac{e^2 \nu_c}{2m_e \omega_M^2}. \tag{61}$$

4.2 DC Discharge Heating

Taking the DC limit $\omega \rightarrow 0$ in the Drude formula, with E_{DC} the (constant) DC discharge field:

$$\lim_{\omega \rightarrow 0} \frac{e^2 \nu_c}{2m_e(\nu_c^2 + \omega^2)} = \frac{e^2 \nu_c}{2m_e \nu_c^2} = \frac{e^2}{2m_e \nu_c}. \tag{62}$$

Therefore:

$$P_{abs}^{(DC)} = \frac{e^2}{2m_e \nu_c} E_{DC}^2 \equiv \alpha_{DC} E_{DC}^2, \quad \alpha_{DC} = \frac{e^2}{2m_e \nu_c}. \tag{63}$$

The ratio $\alpha_{DC}/\alpha_{cav} = \omega_M^2/\nu_c^2 \gg 1$ confirms that DC heating dominates for $\omega_M \gg \nu_c$.

4.3 Electron-Beam Heating

An electron beam interacting with a neutral gas mostly deposits energy into the electron gas through collisional heating, rather than producing free electrons directly. The energised electrons then drive both ionisation and vibrational excitation. The causal chain runs:

$$\boxed{\text{Beam} \rightarrow \varepsilon \rightarrow k_{\text{ion}}, k_{EV} \rightarrow \eta, N \rightarrow \mathcal{E}.} \tag{64}$$

We accordingly represent the beam pulse as a heating term in the electron energy equation. The dimensional energy balance is:

$$\frac{d\varepsilon_{phys}}{dt} = \alpha_{cav} E_{phys}^2 + \alpha_{DC} E_{DC}^2 + \underbrace{P_{\text{beam}}(t)}_{\text{beam heating}} - \frac{\varepsilon_{phys} - \varepsilon_0}{\tau_\varepsilon}, \tag{65}$$

where $P_{\text{beam}}(t) = P_0^{\text{beam}} g(t)$ is the time-dependent power density deposited by the beam, $g(t)$ is the pulse envelope, ε_0 is the thermal equilibrium electron energy the gas would relax to in the absence of any heating, and τ_ε is the electron energy relaxation time.

4.4 Dimensionless Beam Heating Coefficient

Dividing (65) by $\varepsilon_I \omega_M$, the beam term becomes:

$$\frac{P_{\text{beam}}(t)}{\varepsilon_I \omega_M} = \underbrace{\frac{P_0^{\text{beam}}}{\varepsilon_I n_{\text{ref}} \omega_M}}_{H_{\text{beam}}} \cdot P(\tau), \quad (66)$$

where $P(\tau)$ is the dimensionless pulse shape (defined in Section 12) and the dimensionless beam heating coefficient is:

$$H_{\text{beam}} = \frac{P_0^{\text{beam}}}{\varepsilon_I n_{\text{ref}} \omega_M}. \quad (67)$$

H_{beam} measures the deposited beam power density in units of the natural energy scale $\varepsilon_I n_{\text{ref}} \omega_M$. Raising H_{beam} pushes ε transiently above the vibrational threshold $\varepsilon'_{EV} \approx 0.021$ and, at high enough intensity, above the ionisation threshold $\varepsilon = 1$.

4.5 Electron Energy Balance: Steady State without Beam

Setting $d\varepsilon_{\text{phys}}/dt = 0$ in the absence of the pulse and cavity field, and writing $\varepsilon'_0 = \varepsilon_0/\varepsilon_I$, $H_{DC} = \alpha_{DC} E_{DC}^2/(\omega_M \varepsilon_I)$, $\Gamma_\varepsilon = 1/(\omega_M \tau_\varepsilon)$:

$$\varepsilon_{ss} = \varepsilon'_0 + \frac{H_{DC}}{\Gamma_\varepsilon}. \quad (68)$$

4.6 Dimensionless Energy Equation

Dividing (65) by $\varepsilon_I \omega_M$ and collecting all heating channels:

$$\begin{aligned} \frac{1}{\varepsilon_I \omega_M} \frac{d\varepsilon_{\text{phys}}}{dt} &= \frac{\alpha_{\text{cav}} E_{\text{phys}}^2}{\varepsilon_I \omega_M} + \frac{\alpha_{DC} E_{DC}^2}{\varepsilon_I \omega_M} + \frac{P_{\text{beam}}(t)}{\varepsilon_I \omega_M} - \frac{1}{\varepsilon_I \omega_M} \frac{\varepsilon_{\text{phys}} - \varepsilon_0}{\tau_\varepsilon}, \\ \frac{d(\varepsilon_{\text{phys}}/\varepsilon_I)}{d(\omega_M t)} &= \frac{\alpha_{\text{cav}} E_*^2}{\varepsilon_I \omega_M} \frac{E_{\text{phys}}^2}{E_*^2} + \frac{\alpha_{DC} E_{DC}^2}{\varepsilon_I \omega_M} + H_{\text{beam}} P(\tau) - \frac{1}{\omega_M \tau_\varepsilon} \left(\frac{\varepsilon_{\text{phys}}}{\varepsilon_I} - \frac{\varepsilon_0}{\varepsilon_I} \right). \end{aligned} \quad (69)$$

$$\varepsilon' = H_{\text{cav}} \mathcal{E}^2 + H_{DC} + H_{\text{beam}} P(\tau) - \Gamma_\varepsilon (\varepsilon - \varepsilon'_0), \quad (70)$$

where $H_{\text{cav}} = \alpha_{\text{cav}} E_*^2/(\omega_M \varepsilon_I)$. Equation (70) accounts for every source of energy gain in the electron gas – cavity-field Drude heating, DC Drude heating, and beam heating – together with linear relaxation back to the thermal floor ε'_0 . It does not yet account for the energy the electron gas loses each time an electron undergoes an inelastic collision; that correction is added in Section 7 below.

5 Ionisation Rate: Physical Origin

5.1 Electron-Impact Ionisation Cross-Section

When a free electron collides with a CO molecule, ionisation occurs only if the electron's kinetic energy exceeds the ionisation threshold $\varepsilon_I = 14.0$ eV. The empirical cross-section (Lotz/Drawin form for diatomics) is:

$$\sigma_{\text{ion}}(\varepsilon_{\text{phys}}) = \sigma_0 \left(1 - \frac{\varepsilon_I}{\varepsilon_{\text{phys}}} \right)^m \Theta(\varepsilon_{\text{phys}} - \varepsilon_I), \quad (71)$$

where σ_0 is the peak cross-section, $m \approx 1-2$ for CO, and Θ is the Heaviside step function. This form vanishes below threshold, rises continuously above it, and saturates toward σ_0 at high energy.

5.2 Ionisation Rate Coefficient

The rate coefficient K_{ion} ($\text{m}^3 \text{s}^{-1}$) is the cross-section times electron speed $\bar{v}_e$, averaged over the electron energy distribution. For the mean-energy approximation:

$$K_{ion}(\varepsilon_{phys}) = \sigma_{ion}(\varepsilon_{phys}) \cdot \bar{v}_e = K_0 \left(1 - \frac{\varepsilon_I}{\varepsilon_{phys}}\right)^m \Theta(\varepsilon_{phys} - \varepsilon_I), \quad (72)$$

where $K_0 = \sigma_0 \bar{v}_e$ absorbs the electron speed and has units $\text{m}^3 \text{s}^{-1}$.

5.3 Dimensional Electron Density Equation

The electron density evolves from ionisation, loss, recombination, and the DC background source, where δ is the electron loss rate, β_r the electron-ion recombination coefficient, and S_{DC}^{phys} the dimensional DC electron source. The beam pulse enters through the energy equation (70), corrected in Section 7:

$$\frac{dn_e}{dt} = \underbrace{K_{ion}(\varepsilon_{phys})n_g n_e}_{\text{ionisation}} - \underbrace{\delta n_e}_{\text{loss}} - \underbrace{\beta_r n_e^2}_{\text{recombination}} + \underbrace{S_{DC}^{\text{phys}}}_{\text{DC source}}. \quad (73)$$

5.4 Dimensionless Ionisation Rate and Density Equation

In dimensionless units with $\varepsilon = \varepsilon_{phys}/\varepsilon_I$ and $\eta = n_e/n_{ref}$ (where $n_{ref} = \delta/\beta_r$), the ionisation function is:

$$k_{ion}(\varepsilon) = K'_0 \left(1 - \frac{1}{\varepsilon}\right)^m \text{H}(\varepsilon - 1), \quad K'_0 = \frac{K_0 n_g}{\omega_M}. \quad (74)$$

Here $k_{ion}(\varepsilon)$ is the dimensionless ionisation rate coefficient (a function of ε alone); the ionisation rate appearing in the density equation is $k_{ion}(\varepsilon) \eta$, since ionisation is a two-body process between an electron and a neutral.

Note that $\beta'_r = \beta_r n_{ref}/\omega_M = \delta/\omega_M = \delta'$, consistent with n_{ref} being the density where ionisation gain equals linear loss.

Dividing (73) by $n_{ref}\omega_M$:

$$\frac{d(n_e/n_{ref})}{d(\omega_M t)} = \frac{K_{ion}(\varepsilon_{phys})n_g}{\omega_M} \frac{n_e}{n_{ref}} - \frac{\delta}{\omega_M} \frac{n_e}{n_{ref}} - \frac{\beta_r n_{ref}}{\omega_M} \left(\frac{n_e}{n_{ref}}\right)^2 + \frac{S_{DC}^{\text{phys}}}{n_{ref}\omega_M}. \quad (75)$$

$$\boxed{\eta' = k_{ion}(\varepsilon) \eta - \delta' \eta - \beta'_r \eta^2 + S_{DC}}. \quad (76)$$

$P(\tau)$ is absent from this equation as written; it is reinstated as a particle source in Section 10. All beam heating effects enter through ε' in equation (70).

6 Vibrational Excitation Rate Coefficient

6.1 Two Independent Inelastic Channels

In a CO discharge plasma, electrons undergo two physically distinct inelastic collision processes with ground-state CO molecules:

1. Ionisation:



2. Vibrational excitation:



These two processes are independent: they have different thresholds, different cross-sections, and different products. Vibrational excitation begins at energies nearly 50 times lower than ionisation, so electrons can pump the vibrational levels long before significant ionisation sets in, and most vibrational pumping happens in the regime $\varepsilon_{EV} < \varepsilon < 1$ where ionisation is essentially zero. The two rates are therefore treated independently.

6.2 Phenomenological Form of k_{EV}

Consistent with the approach adopted for k_{ion} , the vibrational excitation rate coefficient is:

$$k_{EV}(\varepsilon) = K'_{EV} \left(1 - \frac{\varepsilon'_{EV}}{\varepsilon}\right)^n H(\varepsilon - \varepsilon'_{EV}), \quad (79)$$

where:

- $K'_{EV} = K_{EV}n_g/\omega_M$ is the dimensionless peak rate coefficient,
- $\varepsilon'_{EV} = \varepsilon_{EV}/\varepsilon_I \approx 0.021$ is the dimensionless vibrational threshold,
- n controls how rapidly the rate rises above threshold ($n = 1-2$).

As with k_{ion} , the excitation rate entering the inversion equation (Section 8) is $k_{EV}(\varepsilon)\eta$, the two-body rate coefficient multiplied by the electron density.

6.3 Dimensionless Threshold Value

$$\varepsilon'_{EV} = \frac{\varepsilon_{EV}}{\varepsilon_I} \approx \frac{0.3 \text{ eV}}{14.0 \text{ eV}} \approx 0.021. \quad (80)$$

Process	Physical threshold	Dimensionless ε'
Vibrational excitation	$\approx 0.3 \text{ eV}$	≈ 0.021
Ionisation	14.0 eV	1.0

Table 1: The factor-of-50 separation means vibrational pumping operates efficiently in the regime $0.021 < \varepsilon < 1.0$ where ionisation is negligible.

6.4 Behaviour in Different Energy Regimes

$$\varepsilon < \varepsilon'_{EV} : \quad k_{EV} = 0, \quad k_{ion} = 0 \quad (\text{no inelastic processes}), \quad (81)$$

$$\varepsilon'_{EV} < \varepsilon < 1 : \quad k_{EV} > 0, \quad k_{ion} = 0 \quad (\text{vibrational pumping only}), \quad (82)$$

$$\varepsilon > 1 : \quad k_{EV} > 0, \quad k_{ion} > 0 \quad (\text{both channels active}). \quad (83)$$

The beam-heating term $H_{\text{beam}}P(\tau)$ drives ε into the regime (82) or beyond, activating vibrational pumping (and, at high enough beam intensity, ionisation) for the duration of the pulse.

7 Energy Sinks: Inelastic Collisions Cost Energy

7.1 Physical Motivation

Equation (70) only tracks the energy the electron gas gains from the cavity field, the DC discharge, and the beam. As written it implicitly treats ionisation and vibrational excitation as energetically free, which cannot be right: ionisation is not free. Every time an electron ionises a CO molecule, it must supply at least the ionisation energy ε_I to liberate the bound electron – in dimensionless units this is exactly $\varepsilon_I/\varepsilon_I = 1$ per event. Likewise, every vibrational excitation event removes an amount of energy ε_{EV} (dimensionless $\varepsilon'_{EV} \approx 0.021$) from the colliding electron and deposits it in the CO vibrational manifold, where it later shows up as the pump term for N in Eq. (91).

Because both channels draw from the same electron energy pool ε , the energy equation needs explicit loss terms equal to the rate of each inelastic process times its threshold energy. Physically, ionisation and vibrational excitation move energy out of the optical/DC/beam-heated electron gas and into the plasma and vibrational subsystems, rather than occurring at no energetic cost.

7.2 Corrected Dimensionless Energy Equation

Adding these two sink terms to (70):

$$\boxed{\varepsilon' = H_{cav}\mathcal{E}^2 + H_{DC} + H_{beam}P(\tau) - \Gamma_\varepsilon(\varepsilon - \varepsilon'_0) - k_{ion}(\varepsilon)\eta - k_{EV}(\varepsilon)\eta\varepsilon'_{EV}.} \quad (84)$$

The ionisation sink $k_{ion}(\varepsilon)\eta$ carries an implicit factor of 1 (the dimensionless ionisation threshold), while the vibrational sink $k_{EV}(\varepsilon)\eta\varepsilon'_{EV}$ is weighted by the much smaller threshold $\varepsilon'_{EV} \approx 0.021$: each vibrational quantum removed from the electron gas costs far less energy than each ionisation event. Both sinks use the same rate coefficients $k_{ion}(\varepsilon)$, $k_{EV}(\varepsilon)$ and the same electron density η that drive the η and N equations, so the same inelastic events that populate the plasma and inversion also deplete the electron energy pool – the model is now energetically self-consistent to leading order (elastic collisional heating/cooling of the bulk electron gas is still absorbed into the phenomenological relaxation rate Γ_ε).

The corresponding general steady state (in the absence of the beam pulse) is:

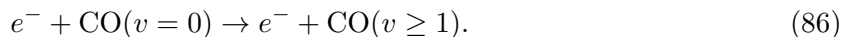
$$\varepsilon_{ss} = \varepsilon'_0 + \frac{H_{cav}\mathcal{E}_{ss}^2 + H_{DC} - k_{ion}(\varepsilon_{ss})\eta_{ss} - k_{EV}(\varepsilon_{ss})\eta_{ss}\varepsilon'_{EV}}{\Gamma_\varepsilon}. \quad (85)$$

The beam heats the electron gas, and the hot electrons then drive ionisation and vibrational excitation through k_{ion} and k_{EV} , with $H_{beam}P(\tau)$ entering only through ε' as a source, while k_{ion} and k_{EV} now also re-enter ε' as sinks.

8 Vibrational Inversion Dynamics

8.1 Physical Origin

In a CO discharge, electrons with energies above $\varepsilon_{EV} \approx 0.3$ eV vibrationally excite ground-state CO molecules:



The vibrationally excited population n_{inv} couples to the cavity field:

$$\text{CO}(v=1) + \gamma_{cav} \rightarrow \text{CO}(v=0) + 2\gamma_{cav}, \quad (87)$$

modifying the cavity dynamics through the phenomenological coupling coefficient κ'_g discussed in Section 11. We define the dimensionless vibrational inversion

$$N = \frac{n_{inv}}{n_{ref}}. \quad (88)$$

The inversion represents the normalized density of vibrationally excited CO molecules produced by electron-impact excitation. It is not an electronic population inversion in the conventional laser sense: it tracks the excess population in the vibrational ladder of the CO ground electronic state relative to the reference density n_{ref} , not a population difference between electronic energy levels.

8.2 Dimensional Inversion Equation

$$\frac{dn_{inv}}{dt} = K_{EV}(\varepsilon_{phys}) n_g n_e - \Gamma_N n_{inv} - \sigma E_{phys}^2 n_{inv}. \quad (89)$$

Here Γ_N is the vibrational relaxation rate (V–T and V–V decay of the excited population back to the ground vibrational state) and σ is the stimulated-emission cross-section coupling the inversion back down through interaction with the cavity field.

Each factor in the pumping term $K_{EV} n_g n_e$ has a clear physical meaning:

- $K_{EV}(\varepsilon_{phys})$: rate at which one electron vibrationally excites one CO molecule ($\text{m}^3 \text{s}^{-1}$);
- n_g : neutral CO number density available for excitation (m^{-3});
- n_e : electron density driving the excitation (m^{-3}).

8.3 Dimensionless Inversion Equation

Dividing (89) by $n_{ref}\omega_M$, with $\Gamma'_N = \Gamma_N/\omega_M$ and $\sigma' = \sigma E_*^2/\omega_M$:

$$\frac{d(n_{inv}/n_{ref})}{d(\omega_M t)} = \frac{K_{EV}(\varepsilon_{phys}) n_g}{\omega_M} \frac{n_e}{n_{ref}} - \frac{\Gamma_N}{\omega_M} \frac{n_{inv}}{n_{ref}} - \frac{\sigma E_*^2}{\omega_M} \frac{E_{phys}^2}{E_*^2} \frac{n_{inv}}{n_{ref}}. \quad (90)$$

$$\boxed{N' = k_{EV}(\varepsilon) \eta - \Gamma'_N N - \sigma' \mathcal{E}^2 N}, \quad (91)$$

where k_{EV} is given by (79) and $\varepsilon'_{EV} \approx 0.021$.

The beam pulse $P(\tau)$ enters only through the energy equation (84). It raises ε , which activates k_{EV} , which then pumps N . Writing $P(\tau)$ directly in the N equation would bypass the electron energy mediation and break the causal chain in (64).

8.4 Saturable Coupling by Adiabatic Elimination

Setting $N' = 0$ in (91):

$$\begin{aligned} k_{EV}(\varepsilon) \eta &= N_{ss}(\Gamma'_N + \sigma' \mathcal{E}^2), \\ N_{ss} &= \frac{k_{EV}(\varepsilon) \eta}{\Gamma'_N + \sigma' \mathcal{E}^2}. \end{aligned} \quad (92)$$

Substituting N_{ss} into the coupling term $\kappa'_g N \mathcal{E}$ in (56):

$$\begin{aligned} \kappa'_g N_{ss} \mathcal{E} &= \frac{\kappa'_g k_{EV}(\varepsilon) \eta}{\Gamma'_N + \sigma' \mathcal{E}^2} \mathcal{E} \\ &= \frac{(\kappa'_g/\Gamma'_N) k_{EV}(\varepsilon) \eta}{1 + (\sigma'/\Gamma'_N) \mathcal{E}^2} \mathcal{E} \\ &= \frac{G_0 k_{EV}(\varepsilon) \eta}{1 + \alpha_s \mathcal{E}^2} \mathcal{E}, \end{aligned} \quad (93)$$

where $G_0 = \kappa'_g/\Gamma'_N$ and $\alpha_s = \sigma'/\Gamma'_N$ as defined in Eq. (9).

This has the form of a saturable coupling, with small-signal coefficient $G_0 = \kappa'_g/\Gamma'_N$ and saturation parameter $\alpha_s = \sigma'/\Gamma'_N$.

This adiabatic elimination (92) holds when Γ'_N is the fastest rate in the N -subsystem. In the opposite limit $\Gamma'_N \ll 1$, N becomes an independent slow variable with timescale Γ'^{-1}_N , which matters for relaxation oscillations. The full 7D system keeps N as a dynamical variable valid in either limit. Section 14 gives the physical reason $\Gamma'_N \ll 1$ holds in practice.

9 Resonance Conditions

9.1 Effective Cavity Frequency and Plasma Tuning

From (52), the effective resonance frequency in dimensionless units is:

$$\frac{\omega_{eff}^2}{\omega_M^2} = \Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{sat}}. \quad (94)$$

As η increases, $\omega_{eff}^2/\omega_M^2$ decreases, but stays bounded below by:

$$\lim_{\eta \rightarrow \infty} \left(\Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{sat}} \right) = \Omega_c^2 + g'^2 - \chi\eta_{sat}. \quad (95)$$

9.2 Plasma Resonance Tuning Condition

Setting $\omega_{eff} = \omega_M$ (cavity resonant with molecular oscillator):

$$\begin{aligned} \Omega_c^2 + g'^2 - \frac{\chi\eta_{res}}{1 + \eta_{res}/\eta_{sat}} &= 1, \\ \frac{\chi\eta_{res}}{1 + \eta_{res}/\eta_{sat}} &= \Omega_c^2 + g'^2 - 1. \end{aligned} \quad (96)$$

Since $g'^2 \ll \Omega_c^2$ in practice:

$$\frac{\chi\eta_{res}}{1 + \eta_{res}/\eta_{sat}} \approx \Omega_c^2 - 1. \quad (97)$$

For a physical solution $\eta_{res} > 0$ we need:

$$\Omega_c^2 + g'^2 > 1 \quad \Rightarrow \quad \Omega_c > \sqrt{1 - g'^2} \approx 1. \quad (98)$$

9.3 Physical Meaning of $\Omega_c > 1$

The cavity (including collective coupling g'^2) has to start above the molecular frequency: $\omega_{eff}(0) > \omega_M$. As electrons build up (η increases), ω_{eff} moves down toward ω_M , bringing the cavity closer to resonance with the molecular source and increasing the coupling and therefore $|\mathcal{E}|$.

9.4 Field Enhancement at Resonance: Heuristic Argument

The argument below is heuristic, meant to illustrate the resonance enhancement effect. In the full 7D system (115), the forcing comes from the coupled nonlinear terms $g'^2 S(x) + \mu_c y + \kappa'_g N \mathcal{E} + E_{seed}$, not from an external drive. The effective forcing amplitude F_0 below stands in for the root-mean-square effect of these internal sources, so it should be read qualitatively rather than quantitatively.

Consider an equivalent driven oscillator at frequency $\omega_{drive} = \omega_M$ (unit dimensionless frequency):

$$\mathcal{Y}' + 2\gamma\mathcal{Y} + \Omega_{eff}^2(\eta)\mathcal{E} = F_0 \cos(\tau), \quad (99)$$

where F_0 represents the effective amplitude of the combined forcing from molecular motion and the inversion-mediated coupling. The steady-state field amplitude is:

$$|\mathcal{E}_{ss}| = \frac{F_0}{\sqrt{\left(\Omega_c^2 + g'^2 - \frac{\chi\eta}{1+\eta/\eta_{sat}} - 1\right)^2 + 4\gamma^2}}. \quad (100)$$

As η increases from 0 toward η_{res} (with $\Omega_c > 1$):

$$\left|\Omega_c^2 + g'^2 - \frac{\chi\eta}{1+\eta/\eta_{sat}} - 1\right| \text{ decreases from } |\Omega_c^2 + g'^2 - 1| > 0 \text{ toward } 0, \quad (101)$$

so $|\mathcal{E}_{ss}|$ increases monotonically – the positive plasma feedback.

For $\eta > \eta_{res}$, the system moves below resonance and the field decreases, so the positive feedback only operates for $0 < \eta < \eta_{res}$.

At $\eta = 0$ (no plasma):

$$|\mathcal{E}_{ss}|_{\eta=0} = \frac{F_0}{\sqrt{(\Omega_c^2 + g'^2 - 1)^2 + 4\gamma^2}}. \quad (102)$$

At $\eta = \eta_{res}$ (exact resonance):

$$|\mathcal{E}_{ss}|_{res} = \frac{F_0}{2\gamma}. \quad (103)$$

The field enhancement factor is:

$$\mathcal{R} = \frac{|\mathcal{E}_{ss}|_{res}}{|\mathcal{E}_{ss}|_{\eta=0}} = \frac{\sqrt{(\Omega_c^2 + g'^2 - 1)^2 + 4\gamma^2}}{2\gamma} \approx \frac{\sqrt{(\Omega_c^2 - 1)^2 + 4\gamma^2}}{2\gamma}. \quad (104)$$

For high- Q cavities ($\gamma \ll |\Omega_c^2 - 1|$):

$$\mathcal{R} \approx \frac{|\Omega_c^2 - 1|}{2\gamma} \gg 1. \quad (105)$$

10 Addition of a Continuous DC Electron Source

10.1 Motivation

The DC discharge contributes to electron heating through the energy equation (84). Beyond heating, a continuously operating DC discharge always maintains a weak background plasma before any external excitation is applied, and the electron density balance should reflect that.

10.2 Modified Electron Density Equation

A constant DC source term S_{DC} (dimensionless DC electron source) is added to the electron density equation, together with the beam source $P(\tau)$:

$$\frac{d\eta}{d\tau} = k_{ion}(\varepsilon)\eta - \delta'\eta - \beta'_r\eta^2 + S_{DC} + P(\tau). \quad (106)$$

10.3 Determination of S_{DC}

Rather than treating S_{DC} as a free parameter, we fix it by requiring that a small background electron density η_{bg} stay in equilibrium in the absence of any pulse or impact ionisation:

$$P(\tau) = 0, \quad k_{\text{ion}} = 0, \quad \frac{d\eta}{d\tau} = 0. \quad (107)$$

Substituting into (106) gives

$$0 = -\delta' \eta_{\text{bg}} - \beta_r' \eta_{\text{bg}}^2 + S_{\text{DC}}, \quad (108)$$

and therefore

$$S_{\text{DC}} = \delta' \eta_{\text{bg}} + \beta_r' \eta_{\text{bg}}^2. \quad (109)$$

In other words, the DC discharge produces electrons at exactly the rate needed to replace those lost to attachment and diffusion (δ') and electron-ion recombination (β_r') at the background level. No new free parameter is introduced: once η_{bg} is chosen, S_{DC} follows directly.

11 Inversion-Mediated Cavity Coupling Coefficient

The coefficient κ_g' appearing in Eq. (56) and Eq. (54) is treated in this work as a free phenomenological parameter. It characterises the strength with which the vibrational inversion N feeds back on the cavity resonance through stimulated emission, but its magnitude is not fixed by a first-principles calculation here. Physically, κ_g' is expected to scale with the cavity quality factor and the strength of the CO $v = 1 \rightarrow 0$ transition, but we do not derive it from those quantities; instead it is specified directly as an input parameter of the model, to be constrained by comparison with data or by sensitivity analysis over the physically plausible range.

The related quantities $G_0 = \kappa_g' / \Gamma_N'$ (small-signal coupling) and $\alpha_s = \sigma' / \Gamma_N'$ (saturation parameter), introduced in Section 8, inherit this phenomenological status.

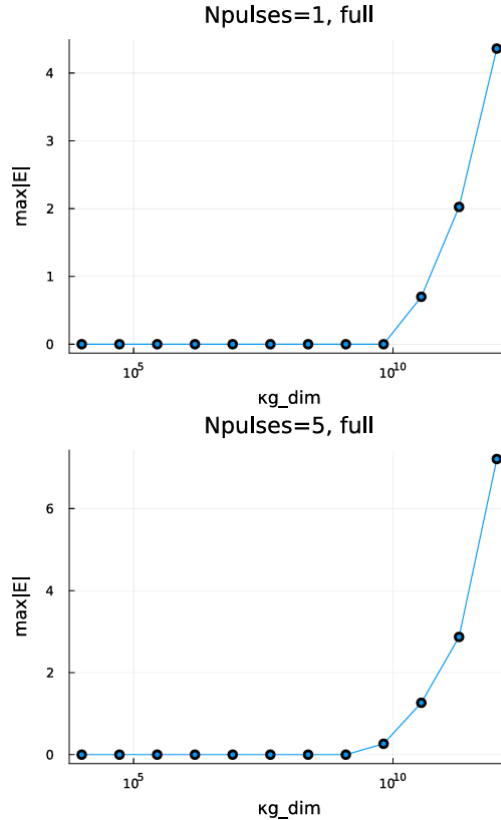


Figure 1: Shows How the field changes with κg

12 Physical Beam Source and Pulse Shape

12.1 The Beam Source

Let $S_{\text{beam}}^{\text{phys}}(t)$ [$\text{m}^{-3} \text{s}^{-1}$] be the volumetric electron production rate due to the electron beam. In dimensional form, the electron balance including all sources is:

$$\frac{dn_e}{dt} = K_{\text{ion}} n_g n_e - \delta n_e - \beta_r n_e^2 + S_{DC}^{\text{phys}} + S_{\text{beam}}^{\text{phys}}(t). \quad (110)$$

This equation is standard in plasma-discharge modelling.

12.2 Non-dimensionalisation

Using $n_e = n_{ref} \eta$ and $\tau = \omega_M t$, dividing (110) by $n_{ref} \omega_M$ gives:

$$\eta' = k_{\text{ion}}(\varepsilon) \eta - \delta' \eta - \beta_r' \eta^2 + S_{DC} + P(\tau), \quad (111)$$

where the dimensionless beam source is:

$$P(\tau) = \frac{S_{\text{beam}}^{\text{phys}}(t)}{n_{ref} \omega_M}. \quad (112)$$

$P(\tau)$ is not an abstract forcing function: it is the volumetric electron production rate of the beam, expressed in units of $n_{ref} \omega_M$.

12.3 Choice of Pulse Profile

Only after defining $P(\tau)$ as the physical beam source do we specify its temporal shape. A Gaussian profile models a beam pulse of finite duration:

$$P(\tau) = P_0 \exp \left[-\frac{(\tau - \tau_0)^2}{2\sigma_b^2} \right], \quad (113)$$

where $P_0 = S_0^{\text{beam}}/(n_{ref} \omega_M)$ is the peak dimensionless injection rate, τ_0 is the pulse arrival time, and σ_b is the pulse duration. The Gaussian is a modelling choice for the beam temporal profile; other profiles (rectangular, triangular) follow by swapping in the appropriate function for the exponential.

12.4 Pulse Train

A train of N_p identical pulses, each separated by the repetition period T , follows directly from (112):

$$P(\tau) = \sum_{n=0}^{N_p-1} P_0 \exp \left[-\frac{(\tau - \tau_0 - nT)^2}{2\sigma_b^2} \right]. \quad (114)$$

Setting $N_p = 1$ recovers the single-pulse model (113); the train is simply the same physical process repeated at intervals T , with no extra derivation needed.

$P(\tau)$ appears in two places with different roles:

1. In the electron density equation (111) as a direct particle source from the beam.
2. In the energy equation (84) multiplied by H_{beam} , representing the thermal energy deposited by the beam electrons.

The ratio H_{beam}/P_0 has units of energy per particle (ε_I), as expected: each injected electron carries kinetic energy that thermalises into the plasma.

13 The Complete 7D System

Incorporating the DC source (109), the beam-heating energy term (67), the independent vibrational excitation rate (79), the seed term (Section 3.8), the energy sinks (Section 7), and the pulse source (114), the complete dimensionless system with state vector $(x, y, \mathcal{E}, \mathcal{V}, \varepsilon, \eta, N)$ is:

$$x' = y, \quad (115a)$$

$$y' = -S(x) - b'y + \mathcal{E}, \quad (115b)$$

$$\mathcal{E}' = \mathcal{V}, \quad (115c)$$

$$\mathcal{V}' = -2\gamma\mathcal{V} - \Omega_{eff}^2(\eta)\mathcal{E} + g'^2 S(x) + \mu_c y + \kappa'_g N \mathcal{E} + E_{seed}, \quad (115d)$$

$$\varepsilon' = H_{cav}\mathcal{E}^2 + H_{DC} + H_{beam} P(\tau) - \Gamma_\varepsilon(\varepsilon - \varepsilon'_0) - k_{ion}(\varepsilon)\eta - k_{EV}(\varepsilon)\eta\varepsilon'_{EV}, \quad (115e)$$

$$\eta' = k_{ion}(\varepsilon)\eta - \delta'\eta - \beta'_r\eta^2 + S_{DC} + P(\tau), \quad (115f)$$

$$N' = k_{EV}(\varepsilon)\eta - \Gamma'_N N - \sigma'\mathcal{E}^2 N, \quad (115g)$$

where $P(\tau)$ is the beam source (114), S_{DC} is given by (109), E_{seed} is the deterministic field seed of Section 3.8, and

$$S(x) = e^{-x} - e^{-2x}, \quad \mu_c = b'g'^2, \quad \text{H} = \text{Heaviside step function}, \quad (116)$$

$$k_{ion}(\varepsilon) = K'_0 \left(1 - \frac{1}{\varepsilon}\right)^{m_I} \text{H}(\varepsilon - 1), \quad (117)$$

$$k_{EV}(\varepsilon) = K'_{EV} \left(1 - \frac{\varepsilon'_{EV}}{\varepsilon}\right)^n \text{H}(\varepsilon - \varepsilon'_{EV}), \quad \varepsilon'_{EV} \approx 0.021, \quad (118)$$

$$\Omega_{eff}^2(\eta) = \Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{sat}}. \quad (119)$$

and the inversion-corrected cavity resonance to be monitored throughout is

$$\Omega_{cav}^2(\tau) = \Omega_{eff}^2(\eta(\tau)) - \kappa'_g N(\tau). \quad (120)$$

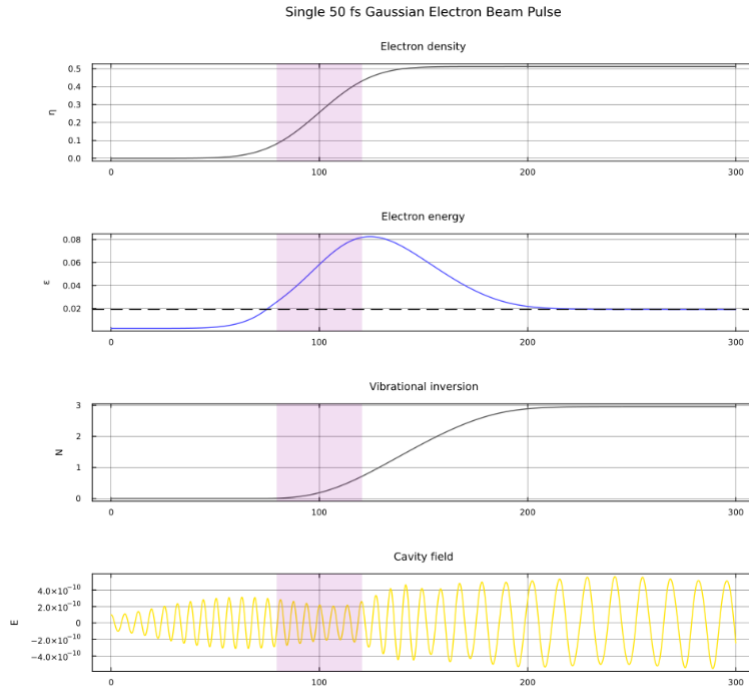


Figure 2: System dynamics at $\kappa g 10^8$

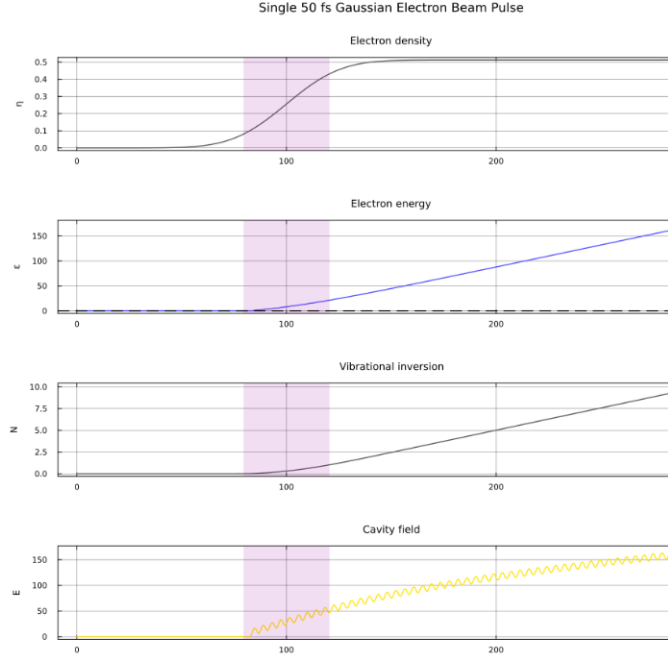


Figure 3: System dynamics at $\kappa g 10^{12}$

phenomenological coupling coefficient introduced in Section 11.

A few structural features of the system are worth pointing out:

- Equation (115d): the seed term E_{seed} regularises the trivial null solution $(\mathcal{E}, \mathcal{Y}) = (0, 0)$ discussed in Section 3.8, without which the deterministic gain terms would have nothing to amplify.
- Equation (115e): the beam pulse $H_{\text{beam}}P(\tau)$ enters the energy equation as a source – the physically correct representation of collisional beam heating – while $k_{\text{ion}}(\varepsilon)\eta$ and $k_{EV}(\varepsilon)\eta\varepsilon'_{EV}$ enter as sinks representing the energetic cost of the inelastic collisions that populate η and N (Section 7).
- Equation (115f): the beam pulse $P(\tau)$ also acts as a particle source, grounded physically as $S_{\text{beam}}^{\text{phys}}/(n_{\text{ref}}\omega_M)$.
- Equation (115g): the pumping term is $k_{EV}(\varepsilon)\eta$, an independently derived vibrational excitation rate rather than a fixed branching ratio of the ionisation rate.
- k_{EV} has threshold $\varepsilon'_{EV} \approx 0.021 \ll 1$, so N grows as soon as electrons are heated above the vibrational excitation threshold, well before ionisation is activated.

14 Timescale Separation

14.1 The Timescales in the Problem

Every dynamical variable in the 7D system relaxes, oscillates, or is driven on its own characteristic dimensionless timescale, and these can be read directly off the linear (diagonal) parts of Eqs. (115a)–(115g):

- The molecular subsystem (x, y) oscillates at the natural frequency of Eq. (26), which is unity by construction: $\tau = \omega_M t$ was chosen precisely so that the Morse oscillator has dimensionless frequency 1. Its timescale is therefore $\mathcal{O}(1)$ in τ , i.e., $\mathcal{O}(\omega_M^{-1})$ in physical time.

- The cavity subsystem $(\mathcal{E}, \mathcal{Y})$ oscillates at $\Omega_c = \mathcal{O}(1)$ (Section ?? gives $\Omega_c \approx 1.05$) and decays at rate 2γ ; both are $\mathcal{O}(1)$ in τ .
- The electron energy ε relaxes at rate $\Gamma_\varepsilon = 1/(\omega_M \tau_\varepsilon)$.
- The electron density η evolves on the ionisation/loss timescale set by K'_0 and δ' .
- The inversion N relaxes at rate $\Gamma'_N = \Gamma_N/\omega_M$.

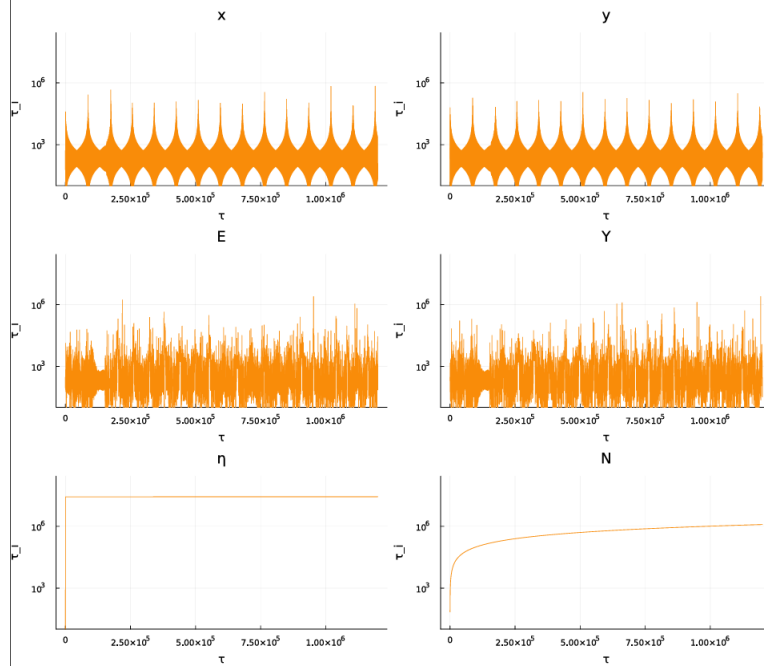


Figure 4: Time Scales

14.2 Numerical Separation of Rates

The key physical distinction between the fast pair (x, y) and the slow variable N comes down to the ratio of their underlying physical rates to ω_M itself. $\omega_M \sim 4 \times 10^{14} \text{ rad s}^{-1}$ (Section ??) is set by the CO bond-stretching motion, one of the fastest mechanical processes available to the molecule: bond vibration happens on sub-picosecond timescales with essentially no competing loss channel that fast, so $b' = b/\omega_M \ll 1$ but the oscillation itself proceeds at the full rate 1 in τ – (x, y) complete an oscillation cycle every $\tau \sim 2\pi$, i.e. every $\sim 10^{-14}$ – 10^{-15} s.

By contrast, Γ_N is a V–T/V–V vibrational relaxation rate: the process by which a vibrationally excited CO molecule ($v = 1$) collisionally relaxes back to $v = 0$. This is a comparatively slow, incoherent, collision-limited process. At typical discharge pressures (~ 1 – 30 Torr) V–T/V–V relaxation times for CO run from microseconds to milliseconds, i.e. $\Gamma_N \sim 10^3$ – 10^6 s^{-1} , whereas $\omega_M \sim 4 \times 10^{14} \text{ s}^{-1}$. Hence

$$\Gamma'_N = \frac{\Gamma_N}{\omega_M} \sim 10^{-11}$$
– $10^{-8} \ll 1, \quad (121)$

many orders of magnitude smaller than b' , γ , or Γ'_ε . The inversion N therefore relaxes on a dimensionless timescale $1/\Gamma'_N \sim 10^8$ – 10^{11} , i.e. it barely moves at all over the $\mathcal{O}(1)$ – $\mathcal{O}(10)$ timescales on which $(x, y, \mathcal{E}, \mathcal{Y})$ complete many oscillation cycles.

14.3 Physical Origin of the Separation

The underlying reason comes down to the type of physical process each variable represents:

- (x, y) describe coherent mechanical motion of a single bond – a restoring force acting directly on a small reduced mass produces a very high natural frequency (ω_M), with essentially no bottleneck.
- N describes a population built up and depleted by incoherent, low-probability collisional events (electron-impact excitation to populate it, V–T/V–V collisions to deplete it). Populating or depleting a population takes many independent collisions, each occurring on the gas-kinetic collision timescale ($\sim n_g \sigma \bar{v}_e$)⁻¹, itself already many orders of magnitude longer than one molecular vibration period; V–T/V–V relaxation is slower still because it typically needs a near-resonant, multi-quantum energy exchange between two heavy vibrational manifolds, an intrinsically inefficient process next to direct mechanical restoring forces.

This is why the 7D system in Eq. (115) is a genuine slow–fast dynamical system, and why the adiabatic elimination of N performed in Section 8 holds whenever Γ'_N is the fastest rate in the N -subsystem specifically (i.e., $\Gamma'_N \gg \sigma' \mathcal{E}^2$ for the fields of interest) even though Γ'_N is by far the slowest rate in the system as a whole.

15 System Dynamics

15.1 Cavity Dynamics

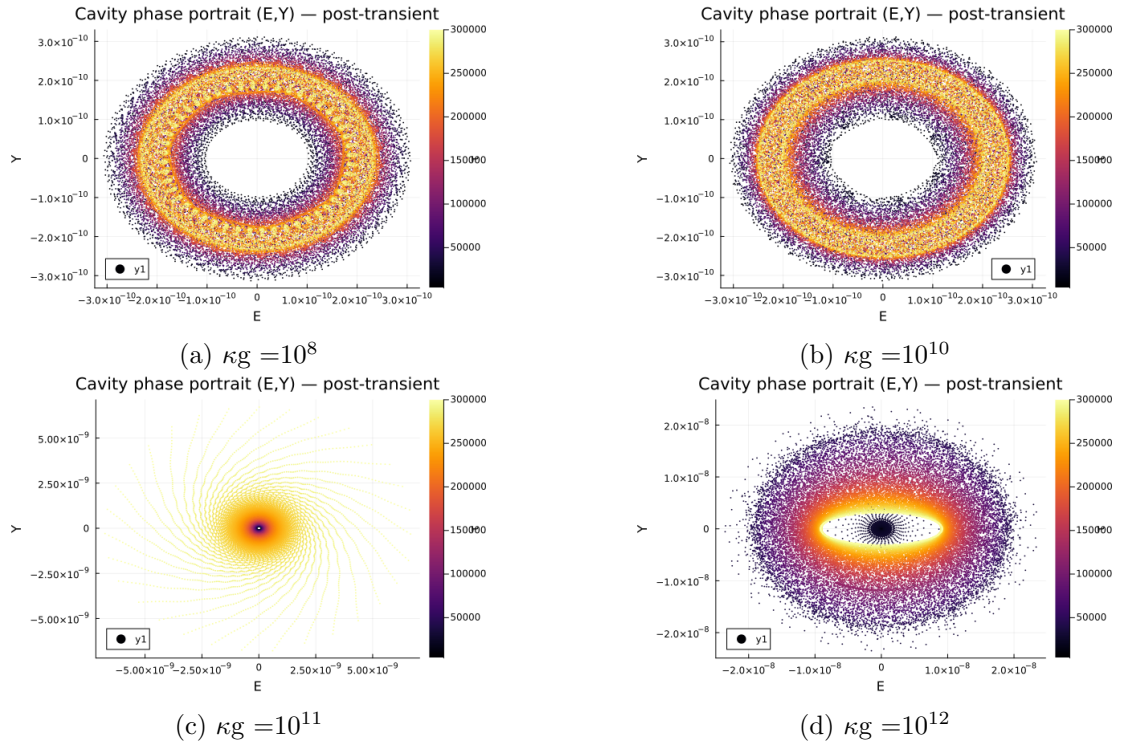


Figure 5: E-Y Phase Portrait for single pulse at different κg values.

Substituting Equation 120 and 119 we get;

$$\Omega_{\text{cav}}^2 = \Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{\text{sat}}} - \kappa'_g N. \quad (122)$$

The cold-cavity resonance is Ω_c^2 . We define the detuning of the actual, dynamically-corrected cavity resonance as:

$$\Delta\Omega_{\text{cav}}^2 \equiv \Omega_{\text{cav}}^2 - \Omega_c^2. \quad (123)$$

or,

$$\Delta\Omega_{\text{cav}}^2 = \left(\Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{\text{sat}}} - \kappa'_g N \right) - \Omega_c^2. \quad (124)$$

or,

$$\Delta\Omega_{\text{cav}}^2 = \Omega_c^2 + g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{\text{sat}}} - \kappa'_g N - \Omega_c^2. \quad (125)$$

or,

$$\boxed{\Delta\Omega_{\text{cav}}^2 = g'^2 - \frac{\chi\eta}{1 + \eta/\eta_{\text{sat}}} - \kappa'_g N.} \quad (126)$$

Physical interpretation: $\Delta\Omega_{\text{cav}}^2$ is defined as the total resonance shift.

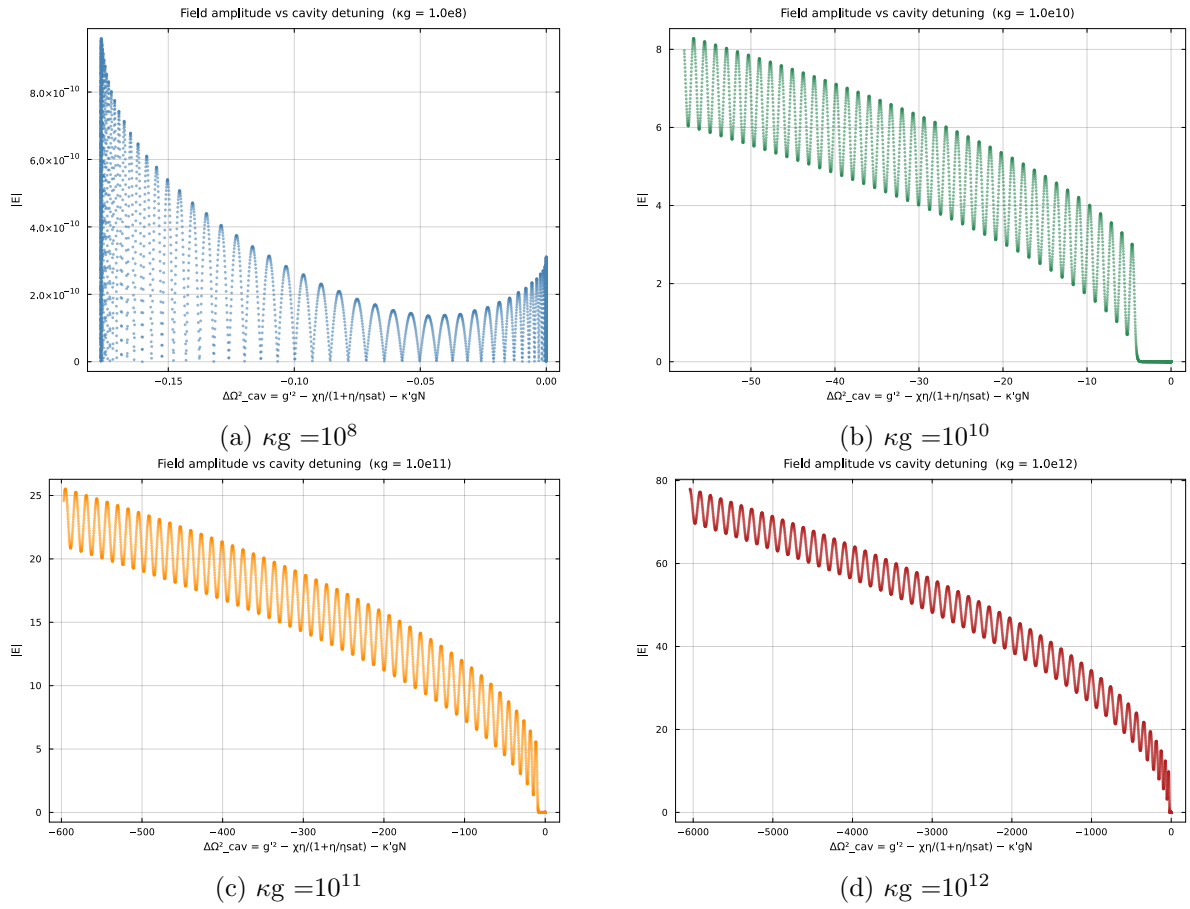
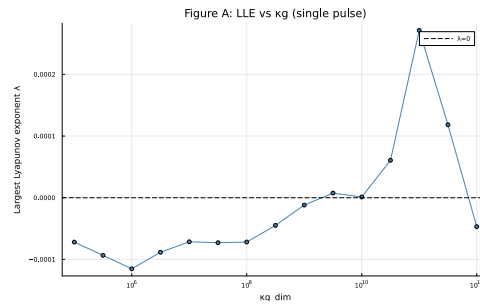


Figure 6: Field Amplitude v/s Cavity Detuning κg values



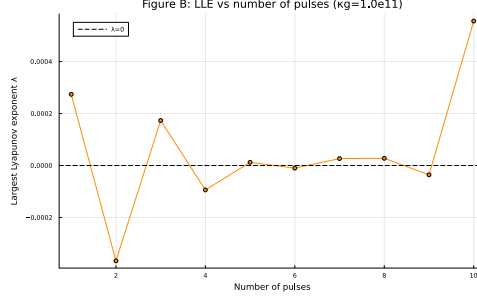


Figure 7: Largest Lyapunov Exponent for N pulses

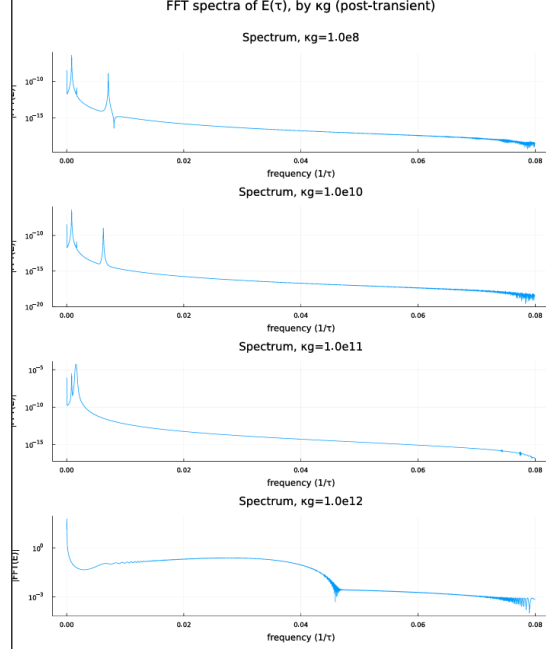
With increasing gain coefficient κ_g , the geometry of the phase portrait (E, Y) undergoes drastic changes due to the imbalance created between cavity gain, cavity loss, and resonance detuning.

The study of $\kappa_g = 10^8$ shows that the largest Lyapunov exponent is negative (but very close to 0) which means that the nearby trajectories will converge. Hence, the dynamics are non-chaotic. The resonance detuning plot correspond to a continuous reduction of the field amplitude, as the cavity progressively returns to its unperturbed resonance. Furthermore, the Fourier spectrum has sharp discrete peaks. As a result, the phase portrait becomes a thick annular torus. The optical field rapidly oscillates near the cavity resonance while slowly modulated in amplitude and phase via the nearly stationary inversion and plasma variables. Because these modulation frequencies do not depend on the optical oscillation, the path never exactly repeats. Instead, it gradually fills a toroidal region of phase space, which is indeed characteristic of quasi-periodic motion.

Rising the gain to $\kappa_g = 10^{10}$ enhances the oscillation amplitude without qualitatively affecting the dynamics. While the gain compensates for the cavity losses, the resonance-detuning curve still exhibits a monotonic decrease of the field amplitude to zero. The LLE is close to zero (or slightly negative) whereas the FFT continues to show spectral peaks. Consequently, the phase portrait continues to be toroidal in shape, with the larger field amplitude merely allowing the trajectory to occupy a larger phase area.

The situation describes a in between. The case $\kappa_g = 10^{11}$. The resonance detuning curve exhibits a monotonically decreasing trend in field values without significant regeneration. Hence, the gain available is still insufficient for self-oscillation. The LLE is also very close to zero, indicating a weakly developed chaos. On the other hand, unlike the lower- κ_g cases, it does not form a torus. In this case, the trajectory spirals inwards toward the equilibrium point giving the spiral structure seen in Fig. 4(c). This thus corresponds to the transient regime between quasi-periodic motion and sustained cavity oscillation.

When $\kappa_g = 10^{12}$, we have a qualitative transition. The resonance–detuning plot does not show a complete decay of the field anymore. Rather, the oscillation amplitude reaches a finite value when the cavity detuning reaches equilibrium. LLE returns close to zero, indicating the existence of a stable periodic attractor. The FFT also changes noticeably, showing a principal oscillation with several nonlinear harmonics associated with gain saturation. Thus, the phase portrait turns into a stable closed elliptical orbit.



15.2 Molecular Dynamics

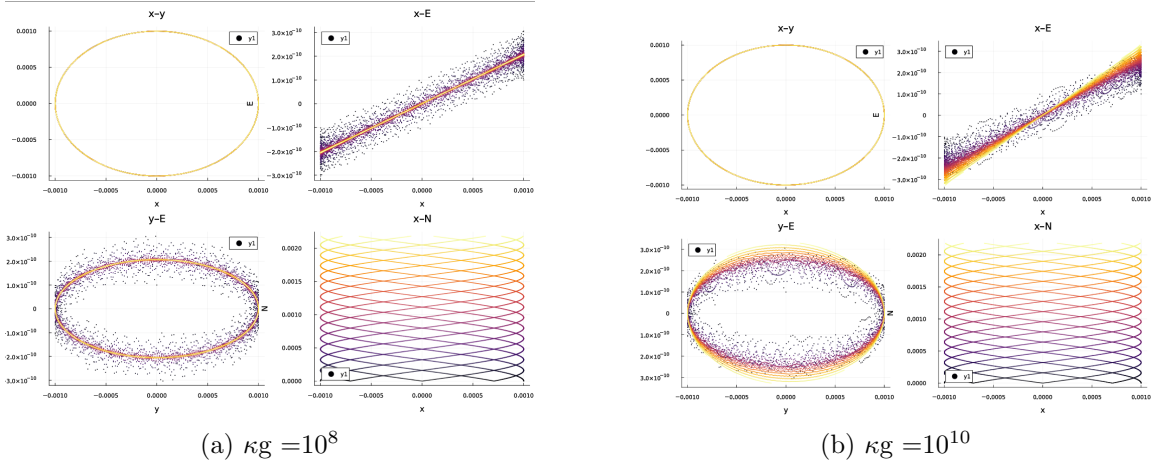


Figure 8: Phase Portraits for single pulse at different κg values.

For $\kappa g = 10^8$:

- x - y : A nice thin ellipse, which is perfectly closed. The amplitude of oscillations of the molecule remains the same during consecutive cycles, there is neither a growth nor a reduction.
- x - E : The points form a straight line, only a few of them are displaced noticeably from it. The field E changes linearly with x , with a slope close to 1, but the amplitude of oscillations of x remains almost the same for different cycles.
- y - E : A similar picture to the previous one, only with y instead of x . The regularities described for the x - y plot are also inherent in this trajectory, and it is seen that y and E also oscillate in phase.
- x - N : A thick, loosely woven lattice in the shape of a braid. The number of oscillations N changes gradually throughout the entire range of x , increasing from 0 to approximately

0.0022 in the given interval. Obviously, the number of oscillations is much larger than the number presented on the plot because x changes rapidly after reaching the maximum amplitude.

For $\kappa_g = 10^{10}$:

- x - y : A cleaner, thinner ellipse of the same shape/amplitude as the $\kappa_g = 10^8$ case. Given that the molecular equation of motion does not directly depend on κ_g but only indirectly through E , this panel does not show significant changes.
- x - E : The central diagonal band is still present, but it is less prominent as there is a noticeable scattered halo around it with a non-negligible density of points near $x \rightarrow \pm 0.001$. The tight correlation seen for $\kappa_g = 10^8$ is now gone.
- y - E : The thin ellipse is still there, but it is now surrounded by a diffuse ring of scattered points, a feature absent in the $\kappa_g = 10^8$ case.
- x - N : The braid is still present but its individual strands are more spaced out and the overall pattern is less tight. The value of N achieves much higher values (~ 0.025 , more than an order of magnitude higher) over the same interval of x , which implies that the oscillations of $N(x)$ as a function of x have a much longer period (the number of oscillations between $x = -0.0025$ and $x = 0.0025$ decreased from ~ 14 to ~ 5). This shows that the ratio between the characteristic timescales of the fast and slow subsystems has decreased.

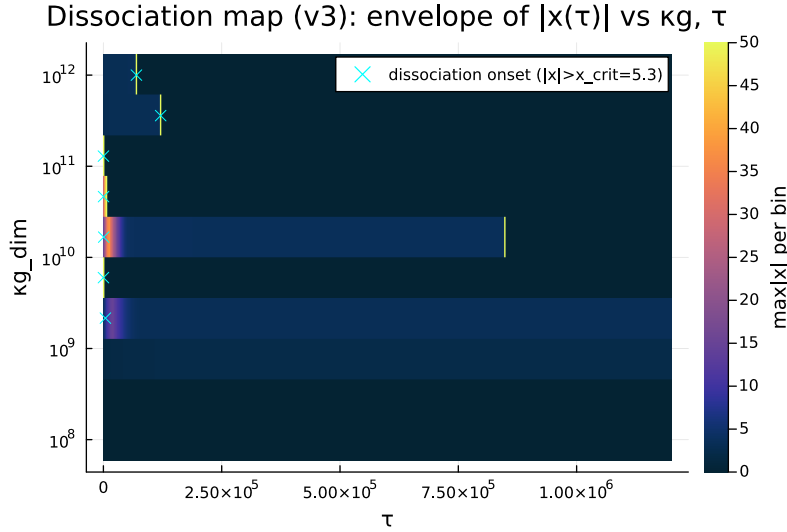


Figure 9: Dissociation Heatmap

The heatmap plots the maximum value of $|x(\tau)|$ reached on each (τ, κ_g) plane. Cyan markers indicate when $|x|$ has first reached $x_{\text{crit}} \approx 5.3$ and dissociation has occurred. The dissociation time rapidly decreases with increasing κ_g : dissociation typically occurs at a much earlier τ for $\kappa_g > \sim 10^{11}$, while $\kappa_g \sim 10^9 - 10^{10}$ leads to only a long $\tau \sim 10^5 - 10^6$ before dissociation. Systems with even smaller κ_g do not dissociate within the plotted range.

16 Contraction Analysis

The Liouville divergence $\nabla \cdot \mathbf{F} = \sum_i \partial F_i / \partial U_i$ is the trace of the Jacobian. Computing each diagonal entry:

$$\frac{\partial x'}{\partial x} = 0, \quad (127)$$

$$\frac{\partial y'}{\partial y} = -b', \quad (128)$$

$$\frac{\partial \mathcal{E}'}{\partial \mathcal{E}} = 0, \quad (129)$$

$$\frac{\partial \mathcal{Y}'}{\partial \mathcal{Y}} = -2\gamma \quad (\text{the coupling term } \kappa'_g N \mathcal{E} \text{ and the seed } E_{\text{seed}} \text{ are independent of } \mathcal{Y}), \quad (130)$$

$$\frac{\partial \varepsilon'}{\partial \varepsilon} = -\Gamma_\varepsilon - k'_{ion}(\varepsilon) \eta - k'_{EV}(\varepsilon) \eta \varepsilon'_{EV}, \quad (131)$$

$$\frac{\partial \eta'}{\partial \eta} = k_{ion}(\varepsilon) - \delta' - 2\beta'_r \eta, \quad (132)$$

$$\frac{\partial N'}{\partial N} = -\Gamma'_N - \sigma' \mathcal{E}^2. \quad (133)$$

The additional terms in (131), $-k'_{ion}(\varepsilon)\eta - k'_{EV}(\varepsilon)\eta\varepsilon'_{EV}$ (with k'_{ion}, k'_{EV} denoting derivatives with respect to ε), come from the energy sinks introduced in Section 7; since both rate coefficients are non-decreasing functions of ε above threshold, these terms are ≤ 0 and so reinforce contraction along ε .

The coupling term $\kappa'_g N \mathcal{E}$ in $\mathcal{Y}'$ contributes to the off-diagonal Jacobian entries $\partial \mathcal{Y}' / \partial \mathcal{E}$ and $\partial \mathcal{Y}' / \partial N$, but not to the diagonal entry $\partial \mathcal{Y}' / \partial \mathcal{Y} = -2\gamma$: the inversion reshapes trajectories within the phase-space volume but does not directly expand it. Note also that S_{DC} , $H_{\text{beam}} P(\tau)$, $P(\tau)$, and E_{seed} are functions of τ only (not of any state variable) and so do not affect any diagonal Jacobian entry.

Summing all diagonal entries:

$$\nabla \cdot \mathbf{F} = -(b' + 2\gamma + \Gamma_\varepsilon + \Gamma'_N + \sigma' \mathcal{E}^2) + k_{ion}(\varepsilon) - \delta' - 2\beta'_r \eta - k'_{ion}(\varepsilon) \eta - k'_{EV}(\varepsilon) \eta \varepsilon'_{EV}. \quad (134)$$

The vibrational inversion N always reinforces phase-space contraction:

$$\nabla \cdot \mathbf{F}|_{7D} = \nabla \cdot \mathbf{F}|_{6D} - \Gamma'_N - \sigma' \mathcal{E}^2 \leq \nabla \cdot \mathbf{F}|_{6D}. \quad (135)$$

In the passive limit, for $\eta = 0$, $N = 0$, $\varepsilon < 1$ (below ionisation threshold, so $k_{ion} = 0$):

$$\nabla \cdot \mathbf{F}|_{\text{passive}} = -(b' + 2\gamma + \Gamma_\varepsilon + \Gamma'_N) < 0. \quad \checkmark \quad (136)$$

Contraction breaks once the active threshold is crossed:

$$k_{ion}(\varepsilon) > b' + 2\gamma + \Gamma_\varepsilon + \Gamma'_N + \sigma' \mathcal{E}^2 + \delta' + 2\beta'_r \eta + k'_{ion}(\varepsilon) \eta + k'_{EV}(\varepsilon) \eta \varepsilon'_{EV}, \quad \varepsilon > 1. \quad (137)$$

17 Threshold Analysis

17.1 Vibrational Pumping Threshold

Inversion N grows as soon as $\varepsilon > \varepsilon'_{EV} \approx 0.021$. From the DC steady-state energy balance (68), this requires:

$$H_{DC} > \Gamma_\varepsilon (\varepsilon'_{EV} - \varepsilon'_0) \approx \Gamma_\varepsilon \times 0.018. \quad (138)$$

This is a weak requirement, satisfied by essentially any operating DC discharge. During the beam pulse, ε is further raised by $H_{\text{beam}} P(\tau)$, subject to the energetic cost of the inelastic sinks in Eq. (84).

17.2 Ionisation Threshold

Both ionisation and plasma growth activate when $\varepsilon > 1$. From the general steady state (85):

$$\varepsilon_{ss} = \varepsilon'_0 + \frac{H_{cav}\mathcal{E}_{ss}^2 + H_{DC} + H_{beam} P(\tau)|_{\text{peak}} - k_{ion}(\varepsilon_{ss})\eta_{ss} - k_{EV}(\varepsilon_{ss})\eta_{ss}\varepsilon'_{EV}}{\Gamma_\varepsilon} > 1. \quad (139)$$

Since $\varepsilon'_0 \ll 1$, this requires approximately:

$$H_{DC} + H_{beam} P(\tau)|_{\text{peak}} > \Gamma_\varepsilon(1 - \varepsilon'_0) + k_{ion}(\varepsilon_{ss})\eta_{ss} + k_{EV}(\varepsilon_{ss})\eta_{ss}\varepsilon'_{EV} \approx \Gamma_\varepsilon. \quad (140)$$

The DC field alone gives $\varepsilon_{ss} \approx H_{DC}/\Gamma_\varepsilon < 1$ at typical discharge conditions; the beam pulse supplies the additional energy needed to cross the threshold, now net of the (generally small) energy siphoned off by the inelastic sinks.

17.3 Density Growth Condition

For η to grow (linearising (115f) at small η , neglecting recombination):

$$\begin{aligned} \eta' &\approx [k_{ion}(\varepsilon) - \delta'] \eta, \\ \text{Growth requires: } k_{ion}(\varepsilon_{ss}) &> \delta', \\ K'_0 \left(1 - \frac{1}{\varepsilon_{ss}}\right)^{m_I} &> \delta'. \end{aligned} \quad (141)$$

17.4 Cavity Resonance Softening

From the adiabatic steady state (92) at small field ($\sigma'\mathcal{E}^2 \ll \Gamma'_N$):

$$\begin{aligned} N_{ss} &\approx \frac{k_{EV}(\varepsilon)\eta}{\Gamma'_N}, \\ \kappa'_g N_{ss} &= G_0 k_{EV}(\varepsilon_{ss})\eta. \end{aligned} \quad (142)$$

As inversion builds up, it progressively softens the effective cavity resonance through the inversion-dependent coupling term: as $\kappa'_g N$ grows, $\Omega_{cav}^2 = \Omega_{eff}^2 - \kappa'_g N$ is reduced relative to Ω_{eff}^2 alone. The electron density η tunes Ω_{eff}^2 through the plasma term, while the inversion N provides a further, independent correction to the cavity response, governed by the phenomenological coupling κ'_g . The quantity

$$\boxed{\Omega_{cav}^2(\tau) = \Omega_{eff}^2(\eta(\tau)) - \kappa'_g N(\tau)} \quad (143)$$

is the natural diagnostic for this behaviour and is evaluated numerically for the full system (115) as a function of H_{beam} , η , N , and the assumed value of κ'_g .

Threshold	Condition	Primary driver
Energy (vib.)	$\varepsilon_{ss} > \varepsilon'_{EV} \approx 0.021$	H_{DC}, H_{beam}
Energy (ion.)	$\varepsilon_{ss} > 1$	H_{DC}, H_{beam}
Density growth	$k_{ion}(\varepsilon_{ss}) > \delta'$	ε_{ss}
Cavity resonance shift	$\Omega_{cav}^2 = \Omega_{eff}^2 - \kappa'_g N$	$\varepsilon_{ss}, \eta, G_0, H_{beam}$
Plasma resonance	$\chi\eta_{res}/(1 + \eta_{res}/\eta_{sat}) = \Omega_c^2 + g'^2 - 1$	Ω_c, χ

Table 2: Threshold hierarchy for the 7D system. All thresholds are controlled through H_{beam} .

18 Conclusion

In conclusion, we have studied a seven-dimensional, non-autonomous dynamical model for the coupled vibro-plasma-cavity-inversion dynamics of electron-beam-pushed CO molecules within an active optical cavity, with the self-consistent inclusion of a Morse-oscillator description of the molecular bond, saturating plasma response, beam-driven electron heating with explicit energy sinks for ionisation and vibrational excitation, a continuous DC electron source, and an effective inversion-mediated cavity coupling κ'_g , governed by the inversion-modified cavity resonance $\Omega_{\text{cav}}^2(\tau) = \Omega_{\text{eff}}^2(\eta(\tau)) - \kappa'_g N(\tau)$. Our numerical exploration of this model has revealed that κ_g drives a transition in the nature of dynamics, as the system progresses from an attracting torus (at low κ_g) via a short transient of a decaying spiral to a stable limit cycle (at high κ_g), while the maximum Lyapunov exponent was found to be close to zero throughout, suggesting that attractors of different dimensionality were encountered along the transitional trajectory. In all cases the increase of κ_g was found to eventually result in dissociation of the molecule due to insufficient restoring force at large bond lengths. In particular, our simulations indicated that the critical stretch $x_{\text{crit}} \approx 5.3$ beyond which the Morse bond is dissociated is approached as κ_g increases. The time-scale for dissociation for a given κ_g decreases precipitously with κ_g , and this value can be seen to determine the upper bound of the region of attraction of the molecular oscillations. As a result of our analysis, we have established that the model displays the characteristic slow-fast structure, with cavity-induced dynamics occurring on a $\mathcal{O}(1)$ time-scale compared to vibrational motion, which is consistent with the vibrational inversion being a collision-dominated process with much longer relaxation time-scale. Our results suggest that the model encapsulates a realistic set of gain-saturation, plasma response and molecular vibrational effects, which can therefore provide a useful test-bed for exploring the attractor structure of such a system in detail. In particular, our findings demonstrate that considerations of the stability of the molecular oscillations at a given set of parameters should be supplemented with an analysis of their dissociation threshold.